

1

result/.../*.png

2

3

3.1 Run 1: piecewise

- : result/251222/exog_sparse_piecewise/images
- : result/251222/exog_sparse_piecewise/images/piecewise_K=1_N=20_T=2000_lambdaS=9.55805_maxweight=0.5_mulambda=0.14386343503607227_num_trials=100_q=20_r=600_rho=0.00440043
- (created_at): 2025-12-22T00:19:11.070470

1: Run 1

PP (Proposed)	ON	$\lambda_S=9.558104 \times 10^{-5}$; $\mu_\lambda=1.438634 \times 10^{-1}$; $q=2.000$
PP-SGD	OFF	$\lambda_S=1 \times 10^{-2}$; $\mu_\lambda=5 \times 10^{-2}$; $q=1$; $r=1$; $\rho=1 \times 10$
PC (Prediction-Correction)	ON	$C=1$; $P=1$; $\alpha=9.502870 \times 10^{-1}$; $\beta=6.953169 \times 10^{-1}$; $\gamma=9$
CO (Correction-Only)	ON	$C=1$; $\alpha=9.502870 \times 10^{-1}$; $\beta_{co}=8.871869 \times 10^{-1}$; $\gamma=9$
SGD	ON	$C=1$; $\alpha=9.502870 \times 10^{-1}$; $\beta_{sgd}=3.536237$; $\lambda_{reg}=6$
PG (Prox-Grad)	OFF	$\lambda_{reg}=1 \times 10^{-3}$; $\max_{iter}=5.000000 \times 10^2$; $step_scale=1.0$

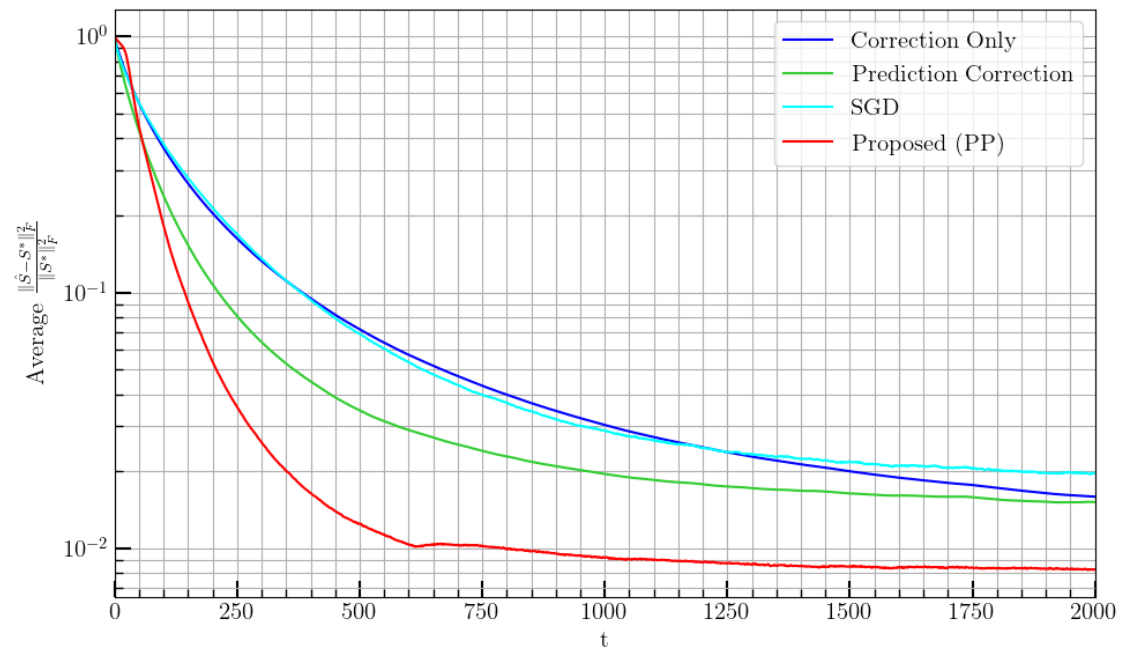
$r, q, \rho, \mu_\lambda, \lambda_S$	PP / PP-SGD
$\lambda_{\text{reg}}, \alpha, \beta, \gamma, P, C$	PC
$\lambda_{\text{reg}}, \alpha, \beta_{\text{co}}, \gamma, C$	CO
$\lambda_{\text{reg}}, \alpha, \beta_{\text{sgd}}, C$	SGD
$\lambda_{\text{reg}}, \text{step_scale}, \text{step_size}, \text{use_fista}, \text{use_backtracking}, \text{max_iter}, \text{tol}$	PG
λ_{offline}	Offline solution (reference)

N ()	$2.000\,000 \times 10^1$
T ()	$2.000\,000 \times 10^3$
K (/)	1
sparsity 0	7×10^{-1}
max_weight	5×10^{-1}
σ_e ()	5×10^{-2}
seed	3
num_trials	$1.000\,000 \times 10^2$

	<code>code.data_gen.generate_piecewise_X_with_exog</code>
S (s_type)	<code>random</code>
T (t_min, t_max)	$(5 \times 10^{-1}, 1.000\,000)$
(z_dist)	<code>uniform01</code>

error_normalization	<code>true_value</code>
divide_by_n2	False
burn_in	−1
burn_in_effective	$6.180\,000 \times 10^2$
offline_lambda_l1	—

(comparison)	
pc_model	<code>exog</code>
pc_use_true_T_init	False
pc_T_init_identity_scale	1.000 000
pp_init_b0	<code>true_T_diag</code>
pp_lookahead	−1
pp_lookahead_effective	$6.180\,000 \times 10^2$



1 piecewise_K=1_N=20_T=2000_lambdaS=9.558103531638434e-
05_maxweight=0.5_mulambda=0.14386343503607227_num_trials=100_q=20_r=600_rho=0.004400494296868914

3.2 Run 2: piecewise

- : result/251226/exog_sparse_piecewise/images
- : result/251226/exog_sparse_piecewise/images/piecewise_K=1_N=20_T=10000_lambdaS=9.55_05_maxweight=0.5_mulambda=0.14386343503607227_num_trials=100_q=20_r=600_rho=0.0044004
- (created_at): 2025-12-26T17:51:24.047664

N ()	$2.000\,000 \times 10^1$
T ()	$1.000\,000 \times 10^4$
K (/)	1
sparsity 0	7×10^{-1}
max_weight	5×10^{-1}
σ_e ()	5×10^{-2}
seed	3
num_trials	$1.000\,000 \times 10^2$

S	(s_type)	<code>code.data_gen.generate_pieewise_X_with_exog</code>
T	(t_min, t_max)	<code>random</code>
	(z_dist)	<code>(5 × 10⁻¹, 1.000 000)</code>
		<code>uniform01</code>

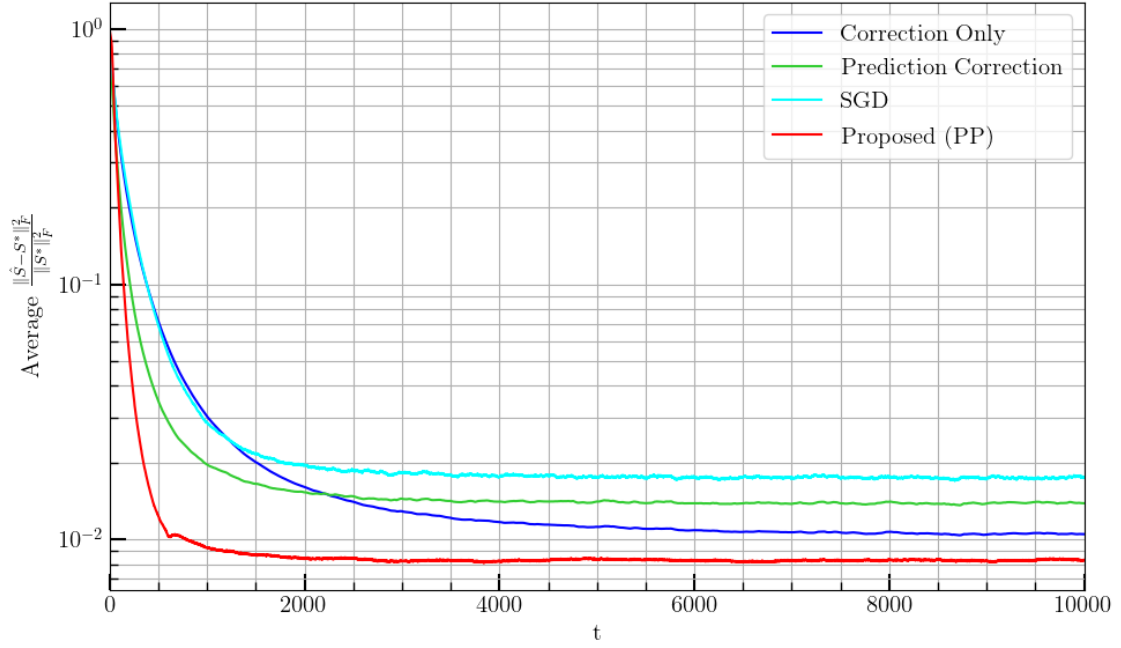
error_normalization	true_value
divide_by_n2	False
burn_in	−1
burn_in_effective	$6.180\,000 \times 10^2$
offline_lambda_l1	—

2: Run 2

PP (Proposed)	ON	$\lambda_S=9.558\,104 \times 10^{-5}$; $\mu_\lambda=1.438\,634 \times 10^{-1}$; $q=2.000$
PP-SGD	OFF	$\lambda_S=1 \times 10^{-2}$; $\mu_\lambda=5 \times 10^{-2}$; $q=1$; $r=1$; $\rho=1 \times 10^{-1}$
PC (Prediction-Correction)	ON	$C=1$; $P=1$; $\alpha=9.502\,870 \times 10^{-1}$; $\beta=6.953\,169 \times 10^{-1}$; γ

CO (Correction-Only)	ON	$C=1$; $\alpha=9.502870 \times 10^{-1}$; $\beta_{co}=8.871869 \times 10^{-1}$; $\gamma=9$
SGD	ON	$C=1$; $\alpha=9.502870 \times 10^{-1}$; $\beta_{sgd}=3.536237$; $\lambda_{reg}=6$
PG (Prox-Grad)	OFF	$\lambda_{reg}=1 \times 10^{-3}$; $\max_iter=5.000000 \times 10^2$; $\text{step_scale}=1.0$

(comparison)	
pc_model	exog
pc_use_true_T_init	False
pc_T_init_identity_scale	1.000 000
pp_init_b0	true_T_diag
pp_lookahead	-1
pp_lookahead_effective	$6.180\,000 \times 10^2$



2 piecewise_K=1_N=20_T=10000_lambdaS=9.558103531638434e-
05_maxweight=0.5_mulambda=0.14386343503607227_num_trials=100_q=20_r=600_rho=0.004400494296868914

3.3 Run 3: linear

- : result/260127/exog_sparse_linear/images
- : result/260127/exog_sparse_linear/images/linear_N=20_T=1000_lambdaS=0.000624284046505_seed=3_stde=0.05_meta.json
- (created_at): 2026-01-27T09:59:38.974995

N ()	$2.000\,000 \times 10^1$
T ()	$1.000\,000 \times 10^3$
sparsity 0	7×10^{-1}
max_weight	5×10^{-1}
σ_e ()	5×10^{-2}
seed	3
num_trials	$1.000\,000 \times 10^2$

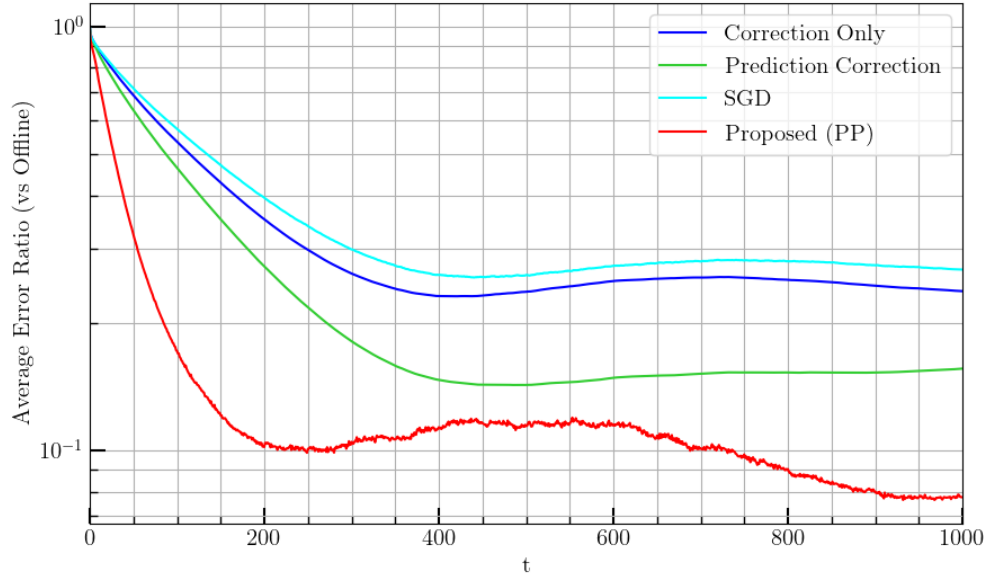
S (s_type)	code.data_gen.generate_linear_X_with_exog random
T (t_min, t_max)	$(5 \times 10^{-1}, 1.000\,000)$
(z_dist)	uniform01

error_normalization	true_value
divide_by_n2	False
burn_in	0
burn_in_effective	—
offline_lambda_l1	—

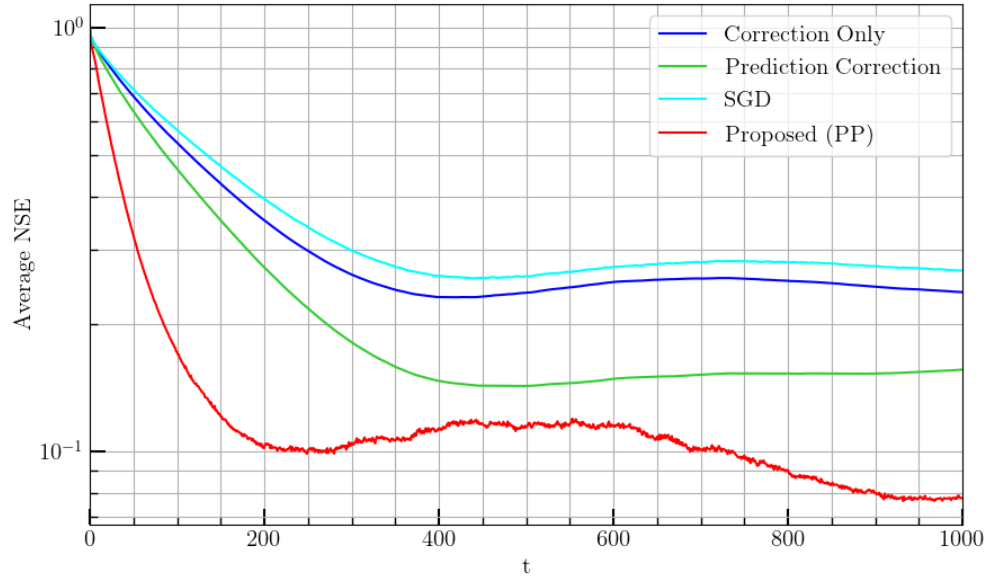
3: Run 3

PP (Proposed)	ON	$\lambda_S=6.242\,840 \times 10^{-4}$; $\mu_\lambda=3.554\,138$; $q=4$; $r=2.000\,0$
PP-SGD	OFF	$\lambda_S=1 \times 10^{-2}$; $\mu_\lambda=1.000\,000$; $q=1$; $r=1$; $\rho=1 \times 10^{-1}$
PC (Prediction-Correction)	ON	$C=5$; $P=2$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta=1.857\,786 \times 10^{-2}$; $\gamma=8$
CO (Correction-Only)	ON	$C=5$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta_{co}=4.048\,497 \times 10^{-5}$; $\gamma=8$

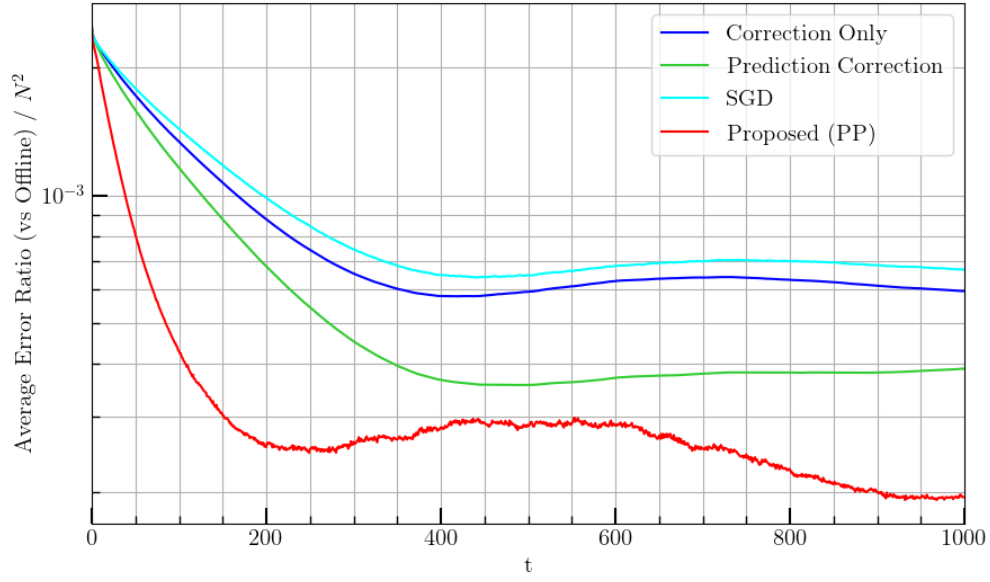
SGD	ON	$C=5$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta_{\text{sgd}}=4.484\,846 \times 10^{-6}$; λ
PG (Prox-Grad)	OFF	$\lambda_{\text{reg}}=1 \times 10^{-3}$; $\max_{\text{iter}}=5.000\,000 \times 10^2$; $\text{step_scale}=1.0$



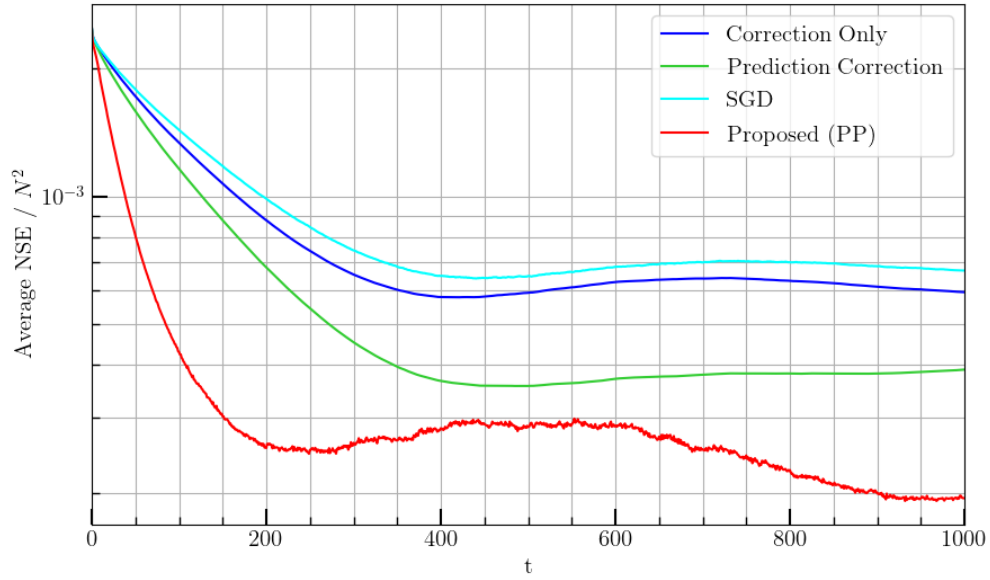
3 linear_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.554137951515271_n2=05_seed=3_stde=0.05.png norm=offline, n2=0



4 linear_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.554137951515271_n2=05_seed=3_stde=0.05.png norm=true, n2=0



5 linear_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.554137951515271_n2=05_seed=3_stde=0.05.png norm=offline, n2=1



6 linear_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.554137951515271_n2=05_seed=3_stde=0.05.png norm=true, n2=1

3.4 Run 4: piecewise

- : result/260127/exog_sparse_piecewise/images
- : result/260127/exog_sparse_piecewise/images/piecewise_K=4_N=20_T=1000_lambdaS=0.00005_seed=3_stde=0.05_meta.json
- (created_at): 2026-01-27T00:25:24.613402

N ()	$2.000\,000 \times 10^1$
T ()	$1.000\,000 \times 10^3$
K (/)	4
sparsity 0	7×10^{-1}
max_weight	5×10^{-1}
σ_e ()	5×10^{-2}
seed	3
num_trials	$1.000\,000 \times 10^2$

S (s_type)	code.data_gen.generate_piecewise_X_with_exog random
T (t_min, t_max)	$(5 \times 10^{-1}, 1.000\,000)$
(z_dist)	uniform01

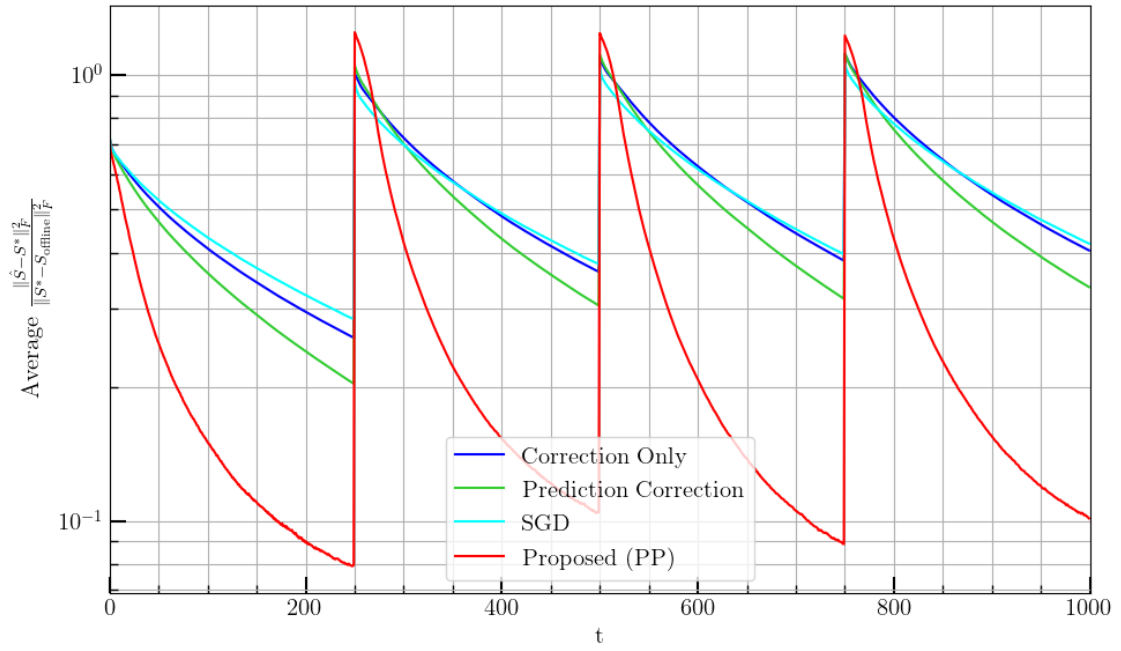
error_normalization	true_value
divide_by_n2	False
burn_in	0
burn_in_effective	0
offline_lambda_l1	$3.162\,278 \times 10^{-2}$

4: Run 4

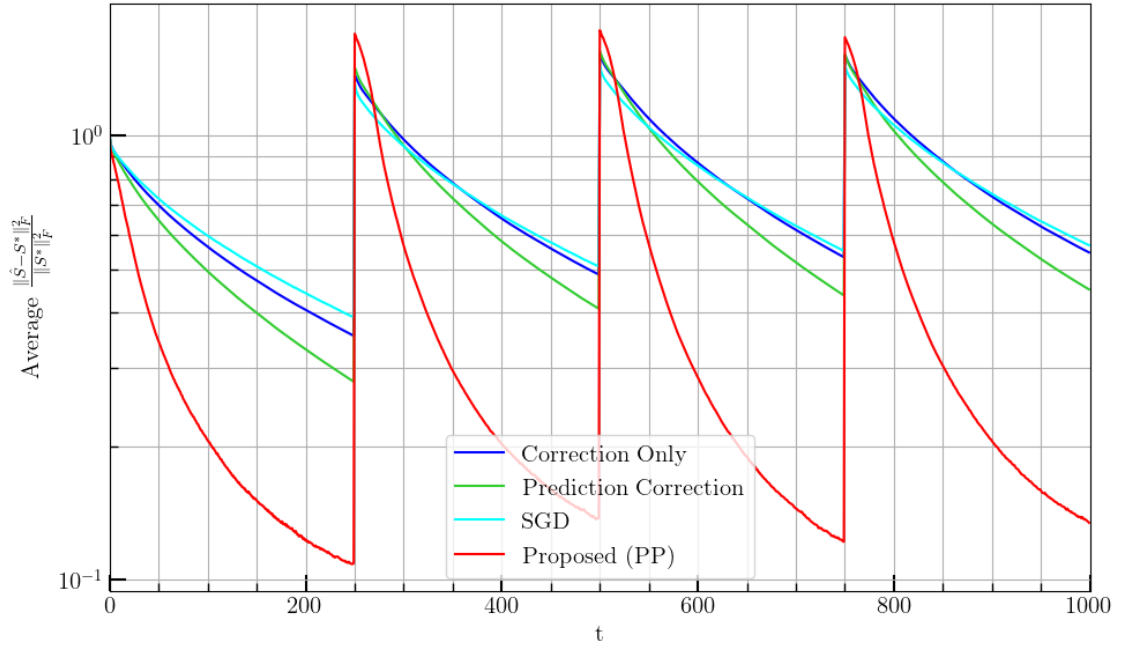
PP (Proposed)	ON	$\lambda_S=6.242\,840 \times 10^{-4}$; $\mu_\lambda=3.554\,138$; $q=4$; $r=2.000\,0$
PP-SGD	OFF	$\lambda_S=1 \times 10^{-2}$; $\mu_\lambda=1.000\,000$; $q=1$; $r=1$; $\rho=1 \times 10$
PC (Prediction-Correction)	ON	$C=5$; $P=2$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta=1.857\,786 \times 10^{-2}$; γ

CO (Correction-Only)	ON	$C=5$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta_{\text{co}}=4.048\,497 \times 10^{-5}$; $\gamma=8$
SGD	ON	$C=5$; $\alpha=7.210\,428 \times 10^{-2}$; $\beta_{\text{sgd}}=4.484\,846 \times 10^{-6}$; λ
PG (Prox-Grad)	OFF	$\lambda_{\text{reg}}=1 \times 10^{-3}$; $\max_{\text{iter}}=5.000\,000 \times 10^2$; $\text{step_scale}=1.0$

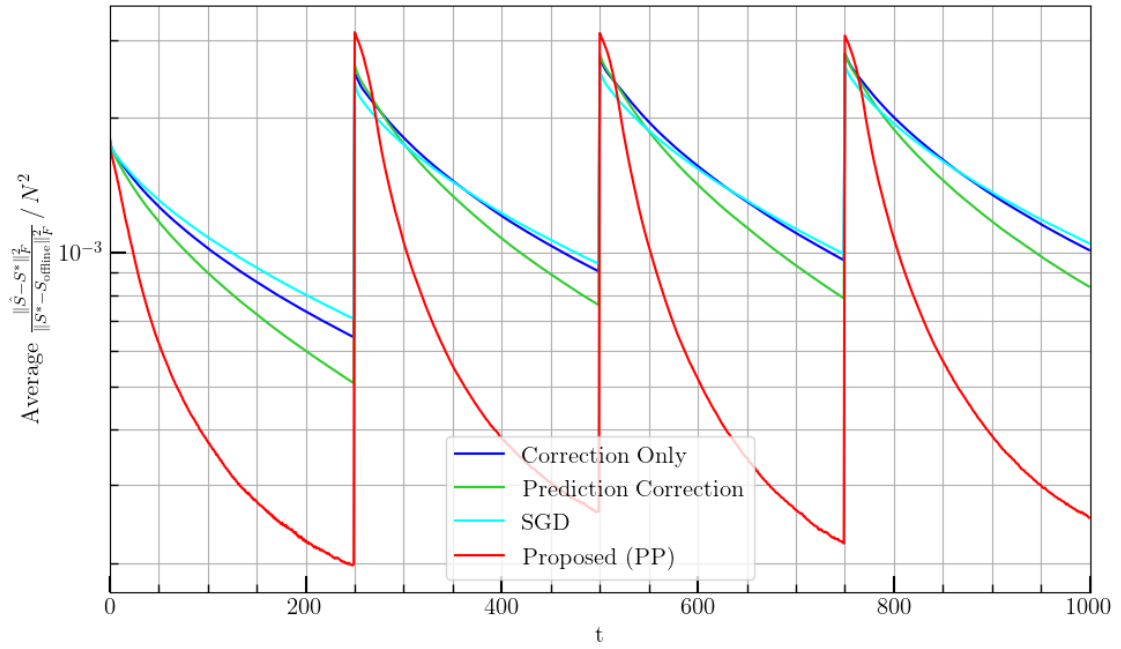
(comparison)	
pc_model	exog
pc_use_true_T_init	False
pc_T_init_identity_scale	1.000 000
pp_init_b0	true_T_diag
pp_lookahead	0
pp_lookahead_effective	0



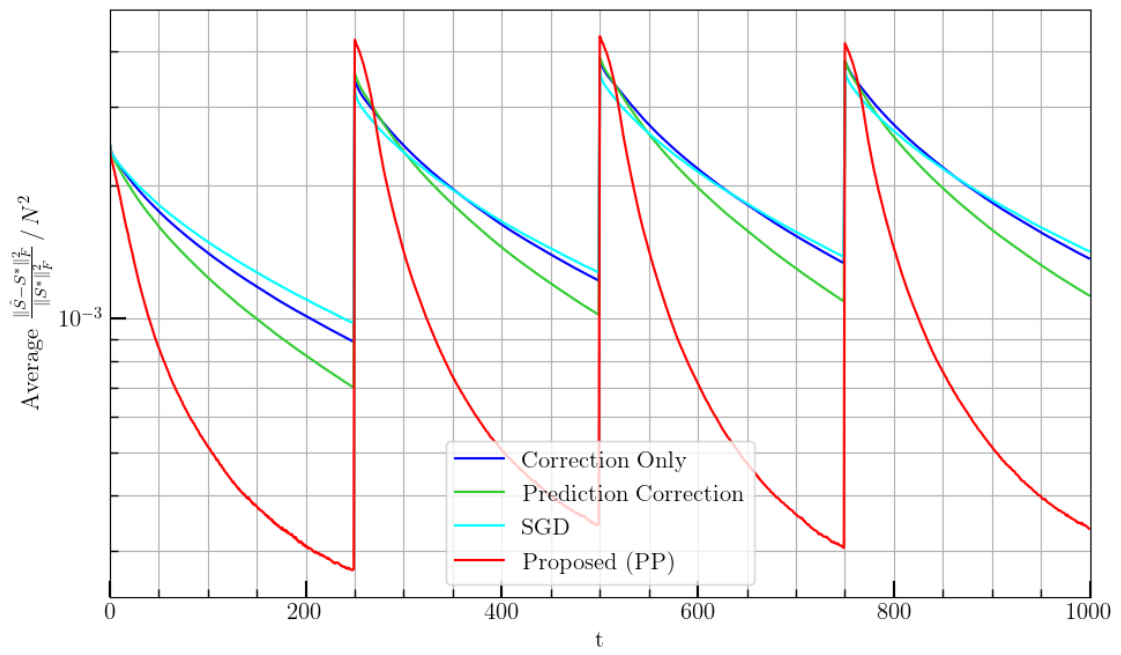
7 piecewise_K=4_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.5541379515105_seed=3_stde=0.05.png norm=offline, n2=0



8 piecewise_K=4_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.5541379515105_seed=3_stde=0.05.png norm=true, n2=0



9 piecewise_K=4_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.5541379515105_seed=3_stde=0.05.png norm=offline, n2=1



10 piecewise_K=4_N=20_T=1000_lambdaS=0.0006242840465350341_maxweight=0.5_mulambda=3.5541379515
05_seed=3_stde=0.05.png norm=true, n2=1